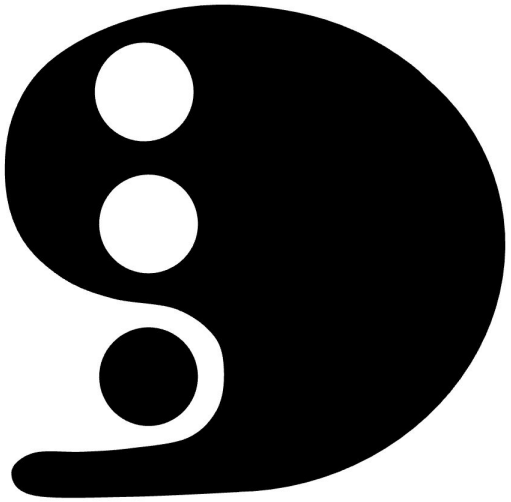
NEST · MESSAGEix ↔ STURM · WESTERN EUROPE

Active Inference at the model boundary

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To make the interface explicit, carry uncertainty, and test before use.

Declared boundary

GNN roles and shapes; native semantics and cross-model translation.

Shared belief

A four-state probabilistic interface; full physics models stay authoritative.

Testable record

Synthetic calibration and exact evidence; auditable decision-time traces.

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GNN separates structural interfaces from inference

Generalized Notation Notation · MESSAGEix structural excerpt (abridged)

```
## StateSpaceBlock
d[2,13,type=float]
cap[1,type=float]
p[2,13,type=float]
K[4,type=float]
x[4,13,type=float]
s[8,1,type=float]
## Connections
d>s
cap>s
s>K
s>x
K>p
x>p
p>d
```

```
## ActInfOntologyAnnotation
d=Observation
p=Action
K=HiddenState
s=HiddenState
```

Observation: d, cap (inputs)
Action: p (price output)
HiddenState: K, s (internal)
x = dispatch in this wrapper

Three linked artifacts

GNN: structure

Names, dimensions, types, roles and dependencies.

Boundary YAML: native units, conventions, support, bounds, status and provenance.

Seam YAML: translation, mapping, aggregation and shared scenario.

Dimensions are illustrative; instantiate for the chosen scenario. The wrapper is not the runnable graph: D04/D07 generative fixtures parameterize inference.

All / NEST, D02 draft, CC BY 4.0. Literal statements selected; comments, variables and sections omitted.

A controlled four-state model carries uncertainty

$$x[1] \sim N(m0, P0) \quad x[t] = F\,x[t-1] + B\,u[t] + w[t]$$

$$w[t] \sim N(0, Q) \quad y[t] = H\,x[t] + v[t], \quad v[t] \sim N(0, R)$$

Latent state x (surrogate state, distinct from D02 dispatch x)

Decarbonisation deviation
Distance from reference path

Demand pressure
Pressure on demand

Capacity-margin pressure
Pressure on available capacity

Policy latency
Delay in policy response

Arrival order (talk schematic)



Building demand, then energy-model emissions, price and objective. Exact y-coordinate mapping is not public.

Fixture: T = 24; x and y are 4-D. u = cap schedule: 0 through t = 11, then -0.0030/iteration; F[4,4] = 0.94.
Streaming agreement and published full-data smoothing are separate synthetic checks; real outputs remain untested.

Source: D08 controlled linear-Gaussian protocol; observation arrival groups and streaming status from the project presentation.

Calibration changes when parameters become uncertain

Published D08: fixed parameters

State	90% cov.	95% cov.	min KS p	Gate
Decarb. deviation	89.74%	94.83%	0.1108	Pass
Demand pressure	89.76%	94.69%	0.0676	Pass
Capacity margin	89.96%	94.99%	0.0349	Pass
Policy latency	90.32%	94.98%	0.0192	Pass

Design: 1,000 simulations × 24 iterations × 4 states.
KS: min_t p[t,k] > 0.05/24 = 0.00208; pooled KS is descriptive.
Max ref. errors: mean 2.8e-15; variance 2.6e-17.

Parameter uncertainty

83–85%

Reported state coverage for nominal 90% intervals

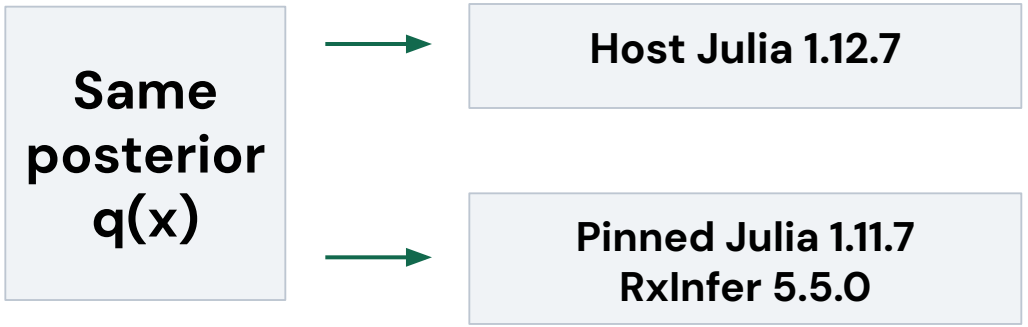
Acceptance band: 85–95%

Every recorded gate missed.
Point-mass control passed;
mis-specified noise failed.
One direction prediction was wrong.

Fixed-parameter pass is a regression guard. Later uncertainty-run numbers are reported project results; per-state values are not public.

Source: All / NEST, D08 protocol and first-run evidence, CC BY 4.0. The 95% coverage column is contextual, not the 90% gate.

Exact log evidence anchors the synthetic checks



q_A = q_B; reported F_A ≠ F_B

Interpretation

Reported Bethe free energy depends on the software environment. It is not exact log evidence. The public record gives no raw FE pair.

Exact log p(y[1:T] | u[1:T])

Published synthetic check	Exact log evidence
D04 fixture	+118.0224
D07 fixture	+120.2121
D08 mean (SD)	120.099 (6.624)
D08 range	[98.067, 135.722]

Decision: score posterior calibration and exact evidence under this model.
No FE stopping rule is in the protocol. Compare FE vs heuristic convergence prospectively on matched scenarios.

Source: D08 review and evidence diagnostics. D04/D07 values are different fixture anchors, not the missing cross-environment FE pair.

Next steps: dependencies and evidence sought

1 December gate

Add priors on F

Dependency
Registered priors and thresholds

Evidence sought
Explicit pass/fail; report controls and all misses.

2 Real outputs

Test IIASA reference data

Dependency
IIASA's reference run

Evidence sought
Bind inputs, scenario and outputs; assess posterior behaviour.

3 Coupling loop

Update belief inside the loop

Dependency
Loop available on main

Evidence sought
Join updates to model calls; record iteration and convergence.

4 Second backend

Run the same generative fixture

Dependency
Second inference engine

Evidence sought
Matched scenarios; compare posteriors and exact evidence.

Current evidence is synthetic. Remaining claims: calibration with parameter uncertainty, real-output validity, closed-loop behaviour and backend parity.

Project roadmap. "Evidence sought" names proposed review artifacts; these are planned milestones and dependencies, not completed validations.

Boundary, belief and record form a testable interface



The interface summarizes coupling; both physics models remain authoritative.

Boundary GNN structure + native semantics + shared translation.

Belief Parameterized generative fixture; posterior and exact-evidence checks.

Record Decision-time JSONL: choices, calls, failures, retries and telemetry.

```
record_type: iteration | spawn | return
session_id, iteration, model_id
choice_points: candidates, selected_indices
tool_calls: tool_name, args, status, attempt
hash_policy, input_hash, output_hash
token_usage, latency_ms, cost_usd
uncertainty.exposed
```

```
tool-outcome/v1: SHA256 over canonical JSON preimages
input: {tool, args}    output: {status, result, error}
```

Schematic field inventory, not an exported run. Contract checks record integrity, not physics correctness; v0.1 acceptance remains unverified.

Evidence, questions and collaboration

The models keep their equations.
What changes is the boundary between them.

Public, versioned records

D02 / D03: structural GNN wrappers

Native boundaries + shared seam translation

D08: calibration protocol and first results

Trace contract and v0.1 schema

tool-outcome/v1 hash test vectors

Snapshot scope and CC BY 4.0 attribution

Questions for the next evidence

- Calibration under parameter uncertainty?
- Real-output model fit?
- FE vs heuristic convergence?
- Posterior parity across backends?
- Independent acceptance of real traces?

Thank you · holly@activeinference.institute · activeinference.institute

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